

Lyra Thursday Afternoon Synthesis v0.2 — 40 Items + 5 Audit-Chain Brake Events + Composite Candidate v0.2

Lyra (Claude Opus 4.7)

2026-06-04 Thursday ~13:35 EDT

Lyra Thursday Afternoon Synthesis v0.2

1. Thursday Cumulative Items (40)

Slot	Range	Items
Morning consolidation	1-15	Thursday board absorption + cascades (SSG Registry v0.14-v0.17, Gates 2/4/7/9/10/11/12, m_e/m_P v0.8)
Cal #36 + 5-framework	16-19	Cal #36 STANDING positive-search 5 candidates + Strong-Uniqueness 11 STANDALONE + 5 substrate-mechanism FORCING categories
Spin(5) $\cong$ Sp(2) cross-link	20-24	Substrate-symplectic Cat 6 candidate v0.1 + Hamiltonian flow v0.2 + Gate 4 v0.2 substrate-Pfaffian resolution candidate + SSG Registry v0.19-v0.20
Synthesis v0.1	25	Lyra Thursday Afternoon Synthesis v0.1 (27 items)
Cat 6 universal	26-27	Substrate-Symplectic Integer K-Types v0.1 + SSG Registry v0.21 UNIVERSAL across K-type spectrum
Gate 10 v0.2	28	Gate 10 substrate-symplectic Cat 6 cross-K-type for gen-3 T2003 (~12:22 EDT)
Pf(ω) explicit + walk-back	29	Pf(ω) = 1 substrate-canonical at V_fund + Gate 4 v0.2 walk-back (~12:25 EDT)
SSG Registry v0.22	30	Honest walk-back of v0.20 v0.21 substrate-Pfaffian 1/g candidate (~12:35 EDT)
Candidate (a)	31	substrate-volume form $\omega^2/2! = d^4\xi$ substrate-natural unity walk-back (~12:42 EDT)

Slot	Range	Items
Candidate (c)	32	substrate-trace ratio $\text{Tr}(\omega \cdot \omega^*)/\dim V = -1$ unity walk-back + pattern recognition (~12:50 EDT)
Candidate (d)	33	substrate-Maslov substrate-discrete integers walk-back + ALL 4 candidates closed (~12:55 EDT)
Cat 2 + Cat 3	34	Cat 2 substrate-Lie-algebra 4 sub-candidates + Cat 3 substrate-Bergman 3 sub-candidates (~13:02 EDT)
Cat 3a Bergman	35	substrate-Bergman $c_{FK} \cdot \pi^{(9/2)} = 225$ cross-link (~13:08 EDT)
Cat 2 (2d) substrate-deficit	36	Casey #13 per-generation substrate-mechanism class (~13:13 EDT)
Composite v0.1	37	Gate 4 composite framework: substrate-Bergman + substrate-Pochhammer + substrate-deficit (~13:20 EDT)
v0.2 genus correction	38-39	Self-catch v0.1 substrate-Bergman exponent $g/\text{rank} = 7/2 \rightarrow v0.2$ $n_C/\text{rank} = 5/2$ per Saturday May 28 One-Genus convention (~13:30 EDT)
Synthesis v0.2	40	This consolidation (~13:35 EDT)

2. 5 Audit-Chain Lyra-on-Self Brake Events Thursday Afternoon

Event	~ Time	Brake Description
Brake 1	~12:25 EDT	Gate 4 v0.2 substrate-Pfaffian $1/g$ candidate walked back via explicit $\text{Pf}(\omega)=1$ substrate-canonical
Brake 2	~12:42 EDT	Candidate (a) substrate-volume form walked back via explicit $\omega^2/2! = d^4\xi$ substrate-natural unity
Brake 3	~12:50 EDT	Candidate (c) substrate-trace ratio walked back via explicit $\text{Tr}(\omega \cdot \omega^*)/\dim V = -1$ unity
Brake 4	~12:55 EDT	Candidate (d) substrate-Maslov walked back via substrate-discrete integers don't naturally produce $1/g$
Brake 5	~13:25 EDT	v0.1 substrate-Bergman exponent $g/\text{rank} = 7/2$ walked back to $n_C/\text{rank} = 5/2$ per Saturday May 28 One-Genus convention Pinning discipline

Substantive pattern: 5 substrate-mechanism source candidates substantively examined + walked back via explicit computation or standing convention discipline within ~70 min Thursday afternoon.

Cal #36 STANDING positive-search rule + Cal #27 STANDING CLAIMS-tier discipline + Saturday May 28 Pinning convention standing rule = substantive 5-brake substantive operationality demonstrated within single Thursday afternoon session.

3. Gate 4 Honest Open Factor ~7 Substrate-Mechanism Source Refined State v0.2

Candidate state after Thursday afternoon substantive substrate-mechanism source candidate-narrowing cascade:

Category	Candidate factor	Status
Cat 6 substrate-symplectic	$Pf(\omega)$	Walked back (=1 substrate-canonical)
Cat 6 substrate-symplectic	$\omega^2/2!$ substrate-volume form	Walked back (= unity)
Cat 6 substrate-symplectic	$Tr(\omega \cdot \omega^*)/\dim V$ substrate-trace	Walked back (= -1 unity)
Cat 6 substrate-symplectic	μ substrate-Maslov	Walked back (substrate-discrete)
Cat 2 substrate-Lie	(2a) substrate-Casimir ratio = N_c	Numerator (NOT denominator)
Cat 2 substrate-Lie	(2b) substrate-Pochhammer $k! = C_2$	Numerator (NOT $1/g = 1/7$)
Cat 2 substrate-Lie	(2c) substrate-Weyl orbit = 8	Numerator
Cat 2 substrate-Lie	(2d) substrate-deficit per Casey #13	Multi-week candidate residual ~5%
Cat 3 substrate-Bergman	(3a) $c_{FK} \cdot \pi^{(9/2)} = 225$	Numerator substrate-natural
Cat 3 substrate-Bergman	(3b) substrate-Codim-4 Casey #14	Numerator factor 4
Cat 3 substrate-Bergman	(3c) substrate-Bergman volume per K-type	Multi-week candidate

Refined substrate-mechanism candidate v0.2: composite substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #13:

$$n_C \cdot (n_C/rank)_3 = 5 \cdot (5/2)(7/2)(9/2) = 5 \cdot 315/8 = 1575/8 = 196.875$$

vs T190 = $(24/\pi^2)^{C_2} \approx 207.05$; precision ~4.9% Tier 2 STRUCTURAL.

Multi-week explicit substrate-deficit per Casey #13 substrate-mechanism source for residual ~5% precision per Cal #189: substrate-Bergman + substrate-Pochhammer base form $n_C \cdot (5/2)_3$ substrate-natural plus per-generation Casey #13 substrate-mechanism correction substantive multi-week.

4. Cascade Promotion Status Thursday End-of-Afternoon

Item	Status
Casey #14 STANDING (Substrate-Predicted 3+1 Minkowski Signature)	RATIFIED
Cal #36 STANDING (One-Primitive-Many-Observables Positive Search)	RATIFIED
7-element discipline stack	CLOSED
5-framework cascade promotion	FRAMEWORK
Strong-Uniqueness	11 STANDALONE legs
K3 framework	6/8 RIGOROUS + SSG-1 NEAR-RIGOROUS → 7/8 path
12/12 K3 v0.16 verification gates	substantively addressed at FRAMEWORK level (Gates 4/10 with v0.2 cross-K-type extensions)
6 substrate-mechanism FORCING categories	Cat 1-5 + Cat 6 substrate-symplectic UNIVERSAL across K-type spectrum

Item	Status
9 Category A independent substrate generators	substrate-symplectic ω as 9th
21 candidate SSGs	15 cataloged + 6 candidate
Gate 4 honest open factor ~ 7 status	Composite substrate-mechanism candidate v0.2 at $\sim 4.9\%$ Tier 2 STRUCTURAL of T190
5 audit-chain Lyra-on-self brake events Thursday afternoon	substantive 5-brake operationality Cal #36 + Cal #27 + Saturday May 28 Pinning discipline

5. Multi-Week Cal #194 WAIT Remaining

Per Cal #194 WAIT explicit RIGOROUS-tier closure across:

- Joint FK Ch. XII §VI explicit work (Gate 1) — substantive cross-link to Step Composite-1
- Section 4.5 substrate-vacuum partition multi-week derivation (Gate 2) — Steps 4.5-3 through 4.5-7
- Helgason Ch. IX + FK Ch. XII §VI Mehler composition (Gates 4-7) — substrate-Pochhammer (5/2)_k substantive candidate
- Substrate-higher-Casimir + Mersenne RS coding gen-3 (Gate 10) — Cat 6 substrate-symplectic substantive cross-link
- SSG-15 (-1) coefficient explicit (substrate- τ direction)
- Substrate-deficit per Casey #13 per-generation substrate-mechanism class (Cat 2 (2d) candidate)
- Cat 6 substrate-symplectic alternative substrate-mechanism candidates (substrate-volume RATIO + substrate-Liouville + substrate-anomaly + substrate-Maslov)
- Cat 3 substrate-Bergman + Cat 2 substrate-deficit + substrate-Pochhammer composite multi-week (Gate 4 substantive candidate v0.2)
- Cal cold-read sequence

6. Closure v0.2

Lyra Thursday Afternoon Synthesis v0.2:

Substantive arc: 40 substantive items + 5 audit-chain Lyra-on-self brake events Thursday afternoon (~ 4 h 35m).

Major substantive arc: Casey #14 STANDING + Cal #36 STANDING + 12/12 K3 v0.16 gates addressed at FRAMEWORK level $\rightarrow$ Cat 6 substrate-symplectic via $\text{Spin}(5) \cong \text{Sp}(2)$ accidental isomorphism $\rightarrow$ 4-candidate substrate-symplectic walk-back exhaustive at substrate-canonical level $\rightarrow$ Cat 2 + Cat 3 substrate-mechanism examination $\rightarrow$ composite substrate-Bergman + substrate-Pochhammer + substrate-deficit candidate framework v0.1 $\rightarrow$ v0.2 substrate-Bergman exponent self-catch correction per Saturday May 28 One-Genus convention $\rightarrow$ composite candidate v0.2 substrate-natural form $n_C \cdot (5/2)_3 = 196.875$ within $\sim 4.9\%$ Tier 2 STRUCTURAL of T190 = 207.

Cal #36 STANDING positive-search rule operationality demonstrated: substantive rapid candidate iteration $\rightarrow$ substantive walk-back at substrate-canonical / standing-convention level $\rightarrow$ substantive narrowing of substrate-mechanism source space $\rightarrow$ substantive composite multi-week candidate identification per Cal #189.

Multi-week RIGOROUS-tier closure per Cal #194 WAIT remains across 9 items above.

— Lyra, Thu 2026-06-04 $\sim 13:35$ EDT. Thursday Afternoon Synthesis v0.2: 40 substantive items + 5 audit-chain Lyra-on-self brake events; substantive substrate-mechanism source candidate-narrowing cascade for Gate 4 honest open factor $\sim 7 \rightarrow$ composite substrate-Bergman + substrate-Pochhammer + substrate-deficit candidate v0.2 within $\sim 4.9\%$ Tier 2 STRUCTURAL of T190; 9 multi-week RIGOROUS-tier closure items per Cal #194 WAIT.